ClO_2 Solution (AFCO 9311)

Safety Data Sheet
according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations
Revision Date: 01/14/2025 Version: 1.2

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY

Product Identifier

Product Form: Mixture
Product Name: ClO_2 Solution
Product Code: AFCO 9311

Intended Use of the Product

Use of the Substance/Mixture: Chlorine dioxide based biocide. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company
AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329
www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Met. Corr. 1	H290
Skin Irrit. 2	H315
Eye Irrit. 2B	H320
Acute Tox. 4 (Inhalation)	H332

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US) :


GHS05


GHS07

Signal Word (GHS-US) : Warning.

Hazard Statements (GHS-US) : H290 - May be corrosive to metals
H315 - Causes skin irritation
H320 - Causes eye irritation
H332 - Harmful if inhaled

Precautionary Statements (GHS-US) : P234 - Keep only in original packaging.
P261 - Avoid breathing fume/gas/mist/vapors/spray.
P264 - Wash exposed areas thoroughly after handling.
P271 - Use only outdoors or in a well-ventilated area.
P280 - Wear protective gloves.
P302+P352 - IF ON SKIN: Wash with plenty of water.
P304+P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P314 - Call a doctor/physician if you feel unwell.
P320 - Specific treatment (see First Aid Measures on this SDS.)
P332+P313 - If skin irritation occurs: Get medical advice/attention.
P337+P313 - If eye irritation persists: Get medical advice/attention.
P362 - Take off contaminated clothing and wash before reuse.
P390 - Absorb spillage to prevent material damage.

P406 - Store in a corrosion resistant container/container with a resistant inner liner.

Other Hazards

Other Hazards Not Contributing to the Classification: Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions.

Unknown Acute Toxicity (GHS-US) Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Mixture

Name	Product identifier	% (w/w)	Classification (GHS-US)
Water	CAS No. 7732-18-5	>99.7%	Not classified
Chlorine dioxide	CAS No. 10049-04-4	0.3%	Acute Tox. 3 (Oral), H301 Skin Corr. 1A, H314 Aquatic Acute 1, H400

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: If symptoms occur, immediately move person away from exposure and into fresh air. Seek immediate medical attention, keep person warm & quiet. If person is not breathing, begin artificial respiration. If breathing is difficult, administer oxygen. Monitor the person closely for delayed development of pulmonary edema, which may occur up to 72 hours after inhalation.

Skin Contact: Concentrated solutions of the material (>1000ppm) may be highly irritating especially on prolonged contact. Remove contaminated clothing immediately. Immediately flush exposed skin with large amounts of water. Wash thoroughly with mild soap. Consult a physician if irritation or burning persists. Contaminated clothing must be laundered before reuse. Lower concentrations (<1000ppm) may cause some irritation with very long exposure.

Eye Contact: If symptoms develop, move patient away from source of exposure and into fresh air. Flush eyes gently with large amounts of water while holding eyelids apart. If symptoms persist or there is any visual difficulty, obtain medical attention.

Ingestion: First aid is not normally required when small amounts of product are ingested. If symptoms develop or if large quantities of product have been ingested, DO NOT induce vomiting. Do not give anything by mouth if the patient is unconscious. Drink large quantities of water. Consult a physician immediately. Neutralization and use of activated charcoal are not recommended.

Most Important Symptoms and Effects Both Acute and Delayed

General: Causes eye and skin irritation. Harmful if inhaled.

Inhalation: Harmful if inhaled.

Skin Contact: Causes skin irritation.

Eye Contact: Causes eye irritation.

Ingestion: Ingestion is likely to be harmful or have adverse effects.

Chronic Symptoms: Not available.

Notes to physician: Probable mucosal damage may contraindicate the use of gastric lavage.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If you feel unwell, seek medical advice (show the label where possible).

SECTION 5: FIREFIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water.

Unsuitable Extinguishing Media: N/A.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive. Chlorine dioxide gas which may evolve from chlorine dioxide solution may spontaneously decompose with a mild energy release at concentrations of 10% in air or greater at standard temperature & pressure. Chlorine dioxide gas may explode with violent force at concentrations of 30% or greater in air at standard temperature & pressure.

Reactivity: Hazardous reactions will not occur under normal conditions.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Use water or water spray to cool containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection. Wear self-contained breathing apparatus with a full-face piece operated in the “positive pressure demand” setting.

Hazardous Combustion Products: May form chlorine, hydrochloric acid gas, oxygen.

Other information: Do not allow run-off from fire fighting to enter drains or water courses.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid prolonged contact with eyes, skin and clothing.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel and persons not wearing protecticve equipment.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper personal protection equipment (PPE).

Emergency Procedures: Ventilate area. Avoid inhalation of vapors & mists.

Environmental Precautions

Prevent entry to sewers and public waters. If run-off occurs, notify proper authories of any runoff as required.

Methods and Material for Containment and Cleaning Up

For Containment: Large spills: Prevent runoff to sewers, streams, lakes, or other bodies of water. **Small spills**: Absorb liquid on vermiculite, floor absorbent, or other absorbent material. Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Methods for Cleaning Up: Stop spill at source. Dike area around spill to prevent spreading, and pump liquid to salvage tank. Remaining liquid may be taken up on sand, clay, earth, vermiculite, floor absorbent, or other absorbent material, and shovel into containers. Flush spill area from which bulk of spill has been removed with water.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: When heated, material emits irritating fumes. In order to prevent the evolution of chlorine dioxide gas into the breathing zones of workers, agitation of the product should be minimized and the product should not be stirred, mixed turbulently, sprayed, or splashed.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store indoors in a dry, cool and well-ventilated place. Store in original container. Keep container closed when not in use. Store at 50-110°F (10-43°C). Do not store outside or in direct sunlight or freezing temperature. Product should not be heated to temperatures above 140°F (60°C). Avoid incompatible materials.

Incompatible Materials: Strong oxidizers, metals, reducing agents, sulfur compounds, carbon monoxide, mercury, organic materials.

Specific End Use(s)

Chlorine dioxide based biocoide. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Chlorine dioxide (10049-04-4)		
USA OSHA	OSHA PEL (TWA) (ppm	0.100 ppm
USA ACGIH	ACGIH TLV (TWA) (ppm)	0.100 ppm
USA ACGIH	ACGIH (STEL) (ppm)	0.300 ppm

The OSHA permissible exposure limit (PEL) for ClO_2 gas in air is 0.1 ppm (0.3 mg/m^3) as an 8-hour time weighted average. NIOSH recommended exposure limits (REL) and ACGIH threshold limit values (TLV) are also 0.1 ppm.

NIOSH and ACGIH short-term exposure limits (STEL) are 0.3 ppm (0.83 mg/m^3) for periods not to exceed 15 minutes. The STEL concentration should not be repeated more than 4 times per day and should be separated by intervals of at least 60 minutes.

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Provide sufficient mechanical ventilation (general and/or local exhaust) to maintain exposure below allowable limits. Ensure all national/local regulations are observed.

Personal Protective Equipment: Protective clothing. Gloves. Protective goggles.

Materials for Protective Clothing: Wear waterproof protective clothing (PVC is preferred) where chlorine dioxide solutions may splash or spray.

Hand Protection: Wear chemically resistant (such as Neoprene) protective gloves.

Eye Protection: Wear splash-proof face & eye protection (PVC is preferred) where chlorine dioxide solution may splash or spray. Safety goggles should be in compliance with OSHA regulations.

Skin and Body Protection: Wear impervious clothing and boots.

Respiratory Protection: Exposure in the workplace should be monitored to determine if worker exposure exceeds the facility – specified exposure “action-level” or the use of product produces adverse health effects or symptoms of exposure. Adequate ventilation should be used to maintain all work areas at concentrations below 0.1 ppm chlorine dioxide, and if generation of vapors or mists is possible, use local ventilation. Where gas concentration may exceed 0.1 ppm chlorine dioxide, only a NIOSH/MSHA approved, half or full-face acid gas respirator should be used. The facility must have a respiratory program that meets requirements established by 29 CFR 1910.134 which includes a program for medical evaluation. For leaks and emergencies where gaseous chlorine dioxide concentrations may exceed 5ppm, a NIOSH/MSHA approved self-contained breathing apparatus, with full face piece, is required.

Other Information: When using, do not eat, drink or smoke.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	:	Liquid
Appearance	:	Clear, yelllow-green
Odor	:	Sharp pungent
Odor Threshold	:	0.1 ppm
pH	:	5 or lower
Relative Evaporation Rate (butylacetate=1)	:	Not available
Melting Point	:	Not available
Freezing Point	:	32°F (0°C)
Boiling Point	:	212°F (100°C)
Flash Point	:	None
Auto-ignition Temperature	:	None
Decomposition Temperature	:	Not available
Flammability (solid, gas)	:	Not available
Lower Flammable Limit	:	Not available

Upper Flammable Limit	:	Not available
Vapor Pressure	:	Not available
Relative Vapor Density at 20°C	:	Not available
Specific Gravity	:	1.00
Solubility	:	Complete
Log Pow	:	Not available
Log Kow	:	Not available
Viscosity, Kinematic	:	Not available
Viscosity, Dynamic	:	Not available
Explosion Data – Sensitivity to Mechanical Impact	:	Not available
Explosion Data – Sensitivity to Static Discharge	:	Not available

SECTION 10: STABILITY AND REACTIVITY

Reactivity: May be corrosive to metals.

Chemical Stability: The product as solution is stable in the dark. On exposure to light, the solution may decompose to an aqueous solution of chloride and chlorate solutions. In regard to vapor (gas) that may evolve from the product, see "Hazardous Decomposition Products" below.

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Heat. Exposure to UV light or sunlight. Agitation.

Incompatible Materials: Strong oxidizers, metals, reducing agents, sulfur compounds, carbon monoxide, mercury, organic materials, phosphorus.

Hazardous Decomposition Products: Chlorine, hydrochloric acid (gas), oxygen.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Likely Routes of Exposure: Ingestion, skin & eye contact, inhalation of vapors which may evolve from product.
Acute Toxicity: Not classified.
LD50 and LC50 Data: Not available.
Skin Corrosion/Irritation: Causes skin irritation.
Serious Eye Damage/Irritation: Causes eye irritation.
Respiratory or Skin Sensitization: Not classified.
Germ Cell Mutagenicity: Not classified.
Teratogenicity: Not available.
Carcinogenicity: Not classified.
Specific Target Organ Toxicity (Repeated Exposure): Not classified.
Reproductive Toxicity: Available information is insufficient to assess risk to the fetus from maternal exposure to this material during pregnancy. Chlorine dioxide did not cause birth defects in laboratory animals even at very high exposure levels.
Specific Target Organ Toxicity (Single Exposure): Not classified.
Aspiration Hazard: Not classified.
Symptoms/Injuries After Inhalation: Irritation, sore throat, wheezing, difficulty breathing.
Symptoms/Injuries After Skin Contact: Redness.
Symptoms/Injuries After Eye Contact: Causes eye irritation and redness.
Symptoms/Injuries After Ingestion: Abdominal pain, nausea, vomiting and diarrhea; it is unlikely to cause serious digestive tract injury. Chlorine dioxide given daily in drinking water at 1-100 ppm caused a decrease in blood glutathione, altered the morphology of erythrocytes, and caused osmotic fragility in laboratory animals.
Potential Chronic Health Effects: Short term exposure: Components in this product may be irritating to the skin, eyes and respiratory tract. Long term exposure: Prolonged inhalation effects may be harmful. In extreme cases, it may cause pulmonary damage and death.

Information on Toxicological Effects

LD50 and LC50 Data:

LC50 Acute Inhalation Rat (mg/L)	2.13 mg/L
LD50 Acute Oral Rat (mg/kg)	>5000 mg/kg

LD50 Acute Dermal Rat (mg/kg)	>5000 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Toxicity – No data available.

Persistence and Degradability

ClO_2 Solution (AFCO 9311)	
Persistence and Degradability	No data available.

Bioaccumulative Potential

ClO_2 Solution (AFCO 9311)	
Bioaccumulative Potential	No data available.

Mobility in Soil No data available.

Other Adverse Effects

Other Information: Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT

Regulated as a hazardous material when shipped by motor vehicle or rail car under UN1760.

Proper Shipping Name : Corrosive Liquid, N.O.S.
Hazard Class 8
Identification number UN1760
Label Codes : 8
Packing Group : III (Must not ship or store in metal containers)
ERG Number 154

SECTION 15: REGULATORY INFORMATION

US Federal Regulations

EPA FIFRA Information

This chemical is a pesticide product registered by the Environmental Protection Agency (EPA Registration Number 75757-2) and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

CAUTION

Causes moderate eye irritation. Avoid contact with eyes, skin, or clothing. Harmful if swallowed, absorbed through the skin, or inhaled. Wash thoroughly with soap and water after handling, and before eating, drinking, chewing gum, using tobacco, or using the toilet.

TSCA (Toxic Substances Control Act) Status - United States

The intentional ingredients of this material are listed.

CERCLA RQ- 40 CFR 302.4(a)

None listed.

SARA 302 Components - 40 CFR 355 Appendix A

None.

SARA 313 Components - 40 CFR 372.65

Section 313 Components	CAS Number	Percent (%)
Chlorine dioxide	10049-04-4	0.3

(Note: the concentration is below the 1.0% de minimis value)

OSHA Process Safety Management 29 CFR 1910

PSM Component(s)	Condition	TQ (lbs.)
Chlorine dioxide		1000

EPA Accidental Release Prevention 40 CFR 68

PSM Component(s)	Condition	TQ (lbs.)
Chlorine Oxide (ClO_2)		1000

International Regulations

Not determined.

US State Regulations: None.

Canadian Regulations

Water (7732-18-5)
Listed on the Canadian DSL (Domestic Substances List) inventory.

Chlorine dioxide (10049-04-4)
Listed on the Canadian DSL (Domestic Substances List) inventory.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date : 01/16/2025

Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

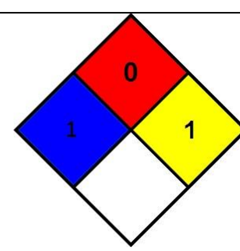
GHS Full Text Phrases:

Acute Tox. 4 (Inhalation)	Acute toxicity (inhalation) Category 4.
Acute Tox. 3 (Oral)	Acute toxicity (oral) Category 3.
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1.
Met. Corr. 1	Corrosive to metals Cat 1.
Eye Irrit. 2B	Serious eye damage/eye irritation Category 2B.
Skin Irrit. 2	Skin corrosion/irritation Category 2.
Skin Corr. 1A	Skin corrosion/irritation Category 1A.
H290	May be corrosive to metals.
H301	Toxic if swallowed.
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H320	Causes eye irritation.
H332	Harmful if inhaled.
H400	Very toxic to aquatic life.

NFPA Health Hazard : 1 - Exposure could cause irritation but only minor residual injury even if no treatment is given.

NFPA Fire Hazard : 0 - Materials that will not burn.

NFPA Reactivity : 1 - Normally stable, but can become unstable at elevated temperatures and pressures or may react with water with some release of energy, but not violently.



HMIS III Rating

Health : 1 Slight Hazard - Irritation or minor reversible injury possible.

Flammability : 0 - Minimal Hazard.

Physical : 1 - Slight Hazard.

Party Responsible for the Preparation of This Document

AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)